SHAKTHI S

shakthi20siva@gmail.com| +91-8610318574 |https://www.linkedin.com/in/shakthi-s-030440249/ |https://github.com/shakthi-20

PROFILE SUMMARY

Electrical Engineering undergraduate with research and project experience in Edge AI, embedded systems, cybersecurity, and machine learning. Published IEEE author with hands-on experience in smart-grid security, IoT systems, and real-time AI deployment.

EDUCATION

Bachelor of Technology, Electrical & Electronics Engineering | Amrita Vishwa Vidyapeetham, Coimbatore | Oct 2022 – Mar 2026 | CGPA: 8.16

Relevant Coursework: Machine Learning, Deep Learning for Visual Computing, Signals and Systems, Digital Signal Processing, AI and Edge Computing, Cybersecurity, IoT Systems, Fundamentals of Soft Computing

Bachelor of Science, Data Science and Applications | IIT Madras, Online | Jan 2023 – Present

Relevant Coursework: Foundation in Data Science and Applications, Machine Learning Foundation, Python Data Structures and Algorithms, Database Management Systems (DBMS), Modern Application Development

RESEARCH & PUBLICATIONS

Breast Cancer Screening from Thermal Images: CNN-Based Analysis with Augmented Datasets | IEEE INSPECT 2025 | Top 60 among 791 applications in AI by Her, AI Impact Summit, 2026 | DOI: 10.1109/INSPECT67393.2025.11350372

Predictive Modeling of Video Streaming Effectiveness Using Artificial Neural Network and Sentiment Analysis | IEEE ICERECT 2025 DOI: 10.1109/ICERECT65215.2025.11375946

Spectrum-Agnostic Space Debris Detection with Autoencoders | IEEE ICAUC 2026 | DOI: 10.1109/ICAUC68182.2026.11441314

PROFESSIONAL EXPERIENCE

Internship Trainee, Satish Dhawan Space Centre SHAR, ISRO | May 2025 – Jun 2025 | Sriharikota, India

Studied Multi-Object Tracking Radar (MOTR), antenna arrays, and launch communication systems used in real-time launch operations at SDSC-SHAR. Prepared technical documentation and workflow diagrams for mission-critical aerospace communication and safety systems.

In-plant Trainee, Kudankulam Nuclear Power Project, NPCIL | Dec 2024 | Kudankulam, India

Conducted diagnostics and maintenance analysis on transformers and circuit breakers under nuclear safety compliance procedures.

KEY RESEARCH PROJECTS

Detection and Mitigation of False Data Injection Attacks on Smart Grid Using Edge AI | *Status: Under Journal Publication*

Engineered real-world, large-scale anomaly detection system for advanced power systems with ESP32 IoT nodes, Raspberry Pi edge computing, and MQTT/TCP-IP telemetry. Simulated 5-class False Data Injection and ARP spoofing attacks in Kali Linux; trained ensemble Machine Learning models (Random Forest, XGBoost, LightGBM, SVM). Deployed real-time Flask/Firebase monitoring dashboards with live edge inference on resource-constrained hardware.

VisionSole — Smart Assistant for Visually Impaired | Github : https://github.com/shakthi-20/VisionSole

Developed voice-controlled navigation system using ESP32-CAM with ~90% real-time object detection accuracy. Integrated ultrasonic sensors for 2-meter obstacle detection range. Built Android application with live path guidance, emergency calling, and SMS alerts, reducing emergency response time by 60% in field trials.

Diabetic Retinopathy Detection — Computer Vision for Healthcare | GitHub: https://github.com/shakthi-20/DiabeticRetinopathy

Developed multi-class CNN using MobileNetV2 transfer learning on retinal fundus images with advanced preprocessing and data augmentation techniques. Achieved 77.63% accuracy on real-world medical imaging dataset, evaluated using Kappa Score and F1-score metrics. Demonstrates practical Deep Learning application to healthcare problems.

TECHNICAL SKILLS

Programming Languages: Python, C, HTML5, CSS

Machine Learning & Deep Learning: TensorFlow, PyTorch, Keras, scikit-learn, XGBoost, LightGBM, Pandas, NumPy

Computer Vision & Signal Processing: CNN, Transfer Learning, MobileNetV2, Autoencoders, Grad-CAM, SHAP

Embedded & Edge AI: ESP32, Arduino, Raspberry Pi, IoT protocols (MQTT), Edge ML inference

Software Engineering: Flask, REST APIs, SQLAlchemy, Git, GitHub

Cloud & Databases: Firebase, MySQL, Cloud integration

Cybersecurity & Tools: Kali Linux, Network attack simulation, VS Code, MATLAB